

Preface

There are no new problems in this file but I just wanted to take the time to compile all of the worksheets we've covered so far into one place. Needless to say, you will not be short on Linear Algebra over the Spring Break.

Also, I just wanted to say thank you to everyone who has shown up, whether it was for 10 minutes or every time, you've made this semester extremely fun!

I hope I have been able to help you all out as we work through this class. If you haven't been able to swing by, just know you're 100% welcome and I'm here to support you!

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1 Week 1: Linear Systems & Invertible Matrices

1.1 Quick Review

This week you covered the basis of every other linear algebra topic you will come across. You covered topics such as linear systems of equations, Gaussian elimination, and matrices and their operations.

A general linear system of equations looks like:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

In matrix form:

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ \vdots & \vdots & \vdots & \vdots \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

Definition 1: Matrix

A matrix is simply an array of numbers, with each number being called an *entry*.

The size of a matrix is defined as (# of rows) $\times$ (# of columns).

If (# of rows) = (# of columns), then the matrix is a *square matrix*.

There are **three** elementary row operations:

1. Scaling: Multiplying *any* row by a scalar.
2. Interchanging: Swapping the position of two rows.
3. Replacement: Adding a constant multiple of one row to another.

These row operations are used in two elimination algorithms:

- Gaussian Elimination $\rightarrow$ Row Echelon Form (REF)
- Gauss-Jordan Elimination $\rightarrow$ Reduced Row Echelon Form (RREF)

Definition 2: Inverse Matrix

The inverse matrix is defined as a matrix such that:

$$AA^{-1} = I, \text{ where } I \text{ is the identity matrix.}$$

A matrix that is invertible is also called *non-singular*.

To find the inverse of a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

1.2 Problems

1. Transform the following linear systems into matrices:

(a)

$$\begin{cases} 5x_1 + 6x_2 = 33 \\ 8x_1 + 9x_2 = 34 \end{cases} \rightarrow \left[\begin{array}{cc|c} & & \\ & & \end{array} \right]$$

(b)

$$\begin{cases} -17x_2 = 33 \\ 81x_1 + 594x_2 = 7231 \end{cases} \rightarrow \left[\begin{array}{cc|c} & & \\ & & \end{array} \right]$$

2. Transform $\mathbf{A}$ into a linear system of equations:

$$\mathbf{A} = \left[\begin{array}{cc|c} 15 & 8 & 1 \\ 9 & -12 & 10 \\ -8 & 10 & 26 \end{array} \right] \rightarrow \begin{cases} x + y = \\ x + y = \\ x + y = \end{cases}$$

3. Determine the size of the coefficient matrices (make sure not to include augmented portions!).

$$\left[\begin{array}{cc|c} a & b & c \\ d & e & f \\ g & h & i \end{array} \right] \quad \left[\begin{array}{cccc|c} 15 & 8 & 4 & 20 & -7 \\ 9 & -12 & 9 & 24 & 23 \end{array} \right] \quad \left[\begin{array}{ccc|c} 12 & 45 & 3 & 1 \\ 13 & -12 & 3 & 10 \\ -8 & 56 & 3 & 26 \\ 4 & 9 & 67 & 15 \\ 4 & 10 & 3 & 28 \\ -8 & 6 & 59 & 88 \end{array} \right]$$

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4. Let $\mathbf{A}$ be the matrix defined below:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ -3 & 7 \end{bmatrix}$$

(a) Find the inverse of $\mathbf{A}$.

(b) What is $\mathbf{A}\mathbf{A}^{-1}$?

$$\mathbf{A}\mathbf{A}^{-1} = \left[\begin{array}{cc} & \\ & \end{array} \right]$$

(c) What is the name of the product of $\mathbf{A}\mathbf{A}^{-1}$?

5. Depending on the progression of your course, you may have encountered 4 to 6 equivalent statements. Throughout the remainder of the course this list will grow **massively**.

(a) What does it mean when we say the statements are equivalent?

- (b) Do the statements apply to *every* matrix?

- (c) The statements are listed in order of implication. List the 6 equivalent statements you have discovered up to now. Use Theorem 1.6.4 in your textbook to find them:

If $\mathbf{A}$ is an _____ matrix, then the following are equivalent.

i. _____

ii. _____

iii. _____

iv. _____

v. _____

vi. _____

6. **(Proof).** Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size. If $\mathbf{AB}$ is nonsingular, prove that $\mathbf{A}$ and $\mathbf{B}$ are invertible.

7. Given the following matrices **A**, **B**, and **C**:

$$\mathbf{A} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 2 & 7 & 8 \\ 6 & -3 & 5 \\ 10 & 1 & 2 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 3 & 6 \\ 5 & 7 & 1 \end{bmatrix},$$

answer the following problems.

(a) Find $\mathbf{A}^{-1}$.

(b) Find $\mathbf{BC}$.

(c) Find $10\mathbf{A} + \mathbf{B}$.

(d) Find $\mathbf{CB}$.

8. Below are two matrices $\mathbf{A}$ and $\mathbf{B}$:

$$\mathbf{A} = \begin{bmatrix} 5 & 3 \\ 1 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

(a) Verify that $\mathbf{A}$ and $\mathbf{B}$ commute ($\mathbf{AB} = \mathbf{BA}$).

(b) Since these two matrices commute, does this mean that *all* matrices commute? If not, why?

(c) (**BONUS**). Can you find a reason why these matrices commute?

2 Week 2: Invertible Matrices, Transformations, & Applications

2.1 Quick Review

This week you covered the tail end of the basis of linear algebra and began working with determinants.

There are four special types of matrices:

Definition 3: Diagonal Matrix

The lower triangular matrix is a matrix where every element of the form $a_{ij} = 0 \forall \{i, j \leq n \mid i \neq j\}$.

$$\mathbf{D} = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$$

Definition 4: Lower & Upper Triangular Matrix

The lower triangular matrix is a matrix where everything *above* the diagonal must be 0.

The upper triangular matrix is a matrix where everything *below* the diagonal must be 0.

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{bmatrix}$$

Definition 5: Symmetric Matrix

A matrix is symmetric when $\mathbf{A} = \mathbf{A}^T$. For example, the following matrix is *symmetric*:

$$\mathbf{D} = \begin{bmatrix} 2 & 0 & 5 & 0 \\ 0 & 6 & 13 & 9 \\ 5 & 13 & 4 & 1 \\ 0 & 9 & 1 & 20 \end{bmatrix}$$

In other words, $a_{ij} = a_{ji}$.

Lastly, linear (or matrix) transformations will come up again and again throughout this course. The two key properties of a linear transformation you will want to remember include:

- **Homogeneity:** $T(k\mathbf{u}) = kT(\mathbf{u})$
- **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$

If you can successfully prove that a transformation follows these, then they must be a linear (or matrix) transformation.

In notation, we represent a transformation T that maps an input from $\mathbb{R}^m$ to an output in $\mathbb{R}^n$ as

$$T : \mathbb{R}^m \rightarrow \mathbb{R}^n$$

If $n = m$, then the transformation is called an operator. Familiar operators are d/dt or $\int$. An operator *does not* have to be linear.

2.2 Problems

1. Use the following systems to (i) Find $\mathbf{A}$, $\vec{x}$, and $\vec{b}$ and (ii) Solve the system using $\mathbf{A}^{-1}$.

(a)

$$\begin{cases} 5x_1 + 6x_2 = 33 \\ 8x_1 + 9x_2 = 34 \end{cases}$$

$$\mathbf{A} = \begin{bmatrix} & \end{bmatrix} \quad \vec{x} = \begin{bmatrix} \\ \end{bmatrix} \quad \vec{b} = \begin{bmatrix} \\ \end{bmatrix}$$

(b)

$$\begin{cases} c_1 t_1 & & & = b_1 \\ & c_2 t_2 & & = b_2 \\ & & c_3 t_3 & = b_3 \\ & & & c_4 t_4 = b_4 \end{cases}$$

$$\mathbf{A} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad \vec{x} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \quad \vec{b} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

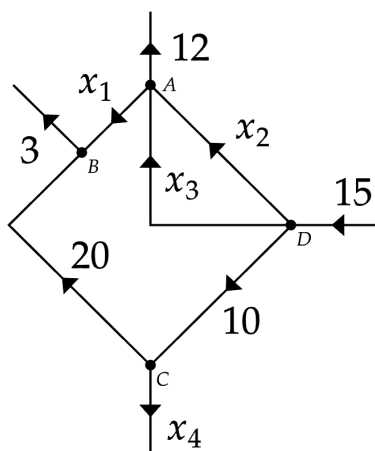
2. Below are a set of conceptual questions.

(a) “ $\mathbf{A}$ is an $m \times n$ diagonal matrix.” What about this statement cannot be right and why?

(b) You state that a linear system can *only* have zero, one, or infinitely many solutions. Your friend, who hasn’t taken linear algebra, disagrees. They say that a parabola can intersect a line twice, meaning there are two solutions for that system and thus, making your statement false. Though his statement is true, why is his claim irrelevant to your statement?

- (c) Your friend just made \$500. He intends to invest in some stocks and purchase some snacks (they are \$5/lb.). Though you may think your friend made some weird choices, you set out to find how many combinations of items your friend can purchase. You create a linear system and find out that the system is (dependent / independent) and (consistent / inconsistent). [Circle the answers in parentheses]
- (d) On what basis did you make your choices for part (c)?

3. Assume that all of the dots (labelled A, B, C, D) must equal 0. Anything entering is positive and anything leaving is negative.



- (a) If one of your x_i turned out to be negative, what would this mean?

- (b) Write a system of equations for the diagram.

$$\left\{ \begin{array}{cccccc} + & + & + & = \\ + & + & + & = \\ + & + & + & = \\ + & + & + & = \end{array} \right.$$

- (c) Solve the system to find x_1, x_2, x_3, x_4 .

- (d) Write your answer from part (c) and write it in vector form.

$$\vec{x} = \begin{bmatrix} \\ \end{bmatrix}$$

4. **(Proof).** If $\mathbf{A}$ is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), prove that $\det(\mathbf{A})$ is the product of the entries on the main diagonal of the matrix; that is, $\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}$.

5. Given the following matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 6 & 10 \\ 6 & -5 & 7 \\ 10 & 4 & -3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 3 & 6 \\ 5 & 7 & 1 \end{bmatrix},$$

answer the following problems.

- (a) Find $\det(\mathbf{A})$.

(b) Find $\det(\mathbf{A}^{-1})$.

(c) Find $\mathbf{A}^{-1}$.

6. Let T be a transformation defined by $T(x_1, x_2) = (x_1 + x_2, 2x_2, 3x_1 - 4x_2)$.

(a) Denote the transformation in the $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ notation.

(b) **(Proof)**. Prove that the transformation T follows the homogeneity property.

(c) **(Proof)**. Prove that the transformation T follows the additivity property.

- (d) Is the transformation T a ...
- i. linear transformation? _____
 - ii. matrix transformation? _____
- (e) Find the standard matrix $\mathbf{A}$ of the transformation T .
- (f) Find $T(6, 7)$.
- (g) If T were to be defined as $T(x_1, x_2) = (x_1^2, x_2^2, 3x_1 - 4x_2)$, why *wouldn't* this be a linear transformation? (I promise this is the last part)

3 Week 3: Determinants & Cramer's Rule

3.1 Quick Review

Over these first four weeks, you got learned the methods of linear algebra that will follow you until the end of the course. For example, you've learned about inverses, matrices, vectors, linear transformations, $\mathbb{R}^n$, and determinants.

You have unlocked 7 out of the 21 (33.3%) equivalent statements you will encounter!

Theorem 1: Invertible Matrix Theorem (IMT)

If $\mathbf{A}$ is an $n \times n$ invertible (nonsingular) matrix, then the following statements are equivalent.

1. $\mathbf{A}$ is invertible.
2. $\mathbf{A}\vec{x} = \vec{0}$ has only the trivial solution.
3. The RREF of $\mathbf{A}$ is $\mathbf{I}_n$.
4. $\mathbf{A}$ can be expressed as a product of elementary matrices.
5. $\mathbf{A}\vec{x} = \vec{b}$ is consistent for every $n \times 1$ matrix $\vec{b}$.
6. $\mathbf{A}\vec{x} = \vec{b}$ has exactly one solution for every $n \times 1$ matrix $\vec{b}$.
7. $\det(\mathbf{A}) \neq 0$.

More recently you learned about how to take determinants. Below are two approaches to finding the determinant.

- You could determine the *cofactors* (C_{ij}) of the determinant and perform the summation as shown below. This is the traditional method.

$$\det(\mathbf{A}) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

- Alternatively, you could evaluate the determinant using row operations as shown below. Although this will work it often takes much longer than cofactor expansion.

If $\mathbf{A}$ is the original $n \times n$ matrix and $\mathbf{B}$ is the resultant matrix (after you perform the operation) then the determinants are related by:

- $\det(\mathbf{B}) = k \det(\mathbf{A})$, $k \in \mathbb{R}^n$ for a **scaling** operation.
- $\det(\mathbf{B}) = -\det(\mathbf{A})$ for an **interchange** operation.
- $\det(\mathbf{B}) = \det(\mathbf{A})$ for a **replacement** operation (adding a scalar multiple of one row to another row).

3.2 Problems

1. Given the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$:

$$\mathbf{A} = \begin{bmatrix} 2 & 7 & 0 & 2 \\ 2 & 0 & 0 & 6 \\ 5 & 3 & 0 & 7 \\ 1 & 0 & 10 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \\ 1 & 5 & 3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 0 & 0 & 6 \\ 3 & 0 & 4 \\ 0 & 10 & 3 \end{bmatrix},$$

solve the following problems.

(a) Find $\mathbf{A}^T$.

(b) Find $\det(\mathbf{A})$.

(c) Find $\mathbf{B}^{-1}$.

(d) Find $\det(\mathbf{B}^{-1})$.

(e) Find $\det(\mathbf{BC})$.

2. Create two matrices $\mathbf{A}$ and $\mathbf{B}$ that satisfy the following condition: $\mathbf{AB} = \mathbf{BA}$, where $\mathbf{A} \neq \mathbf{B} \neq \mathbf{I}$.

3. **(Proof).** Suppose there are two diagonal matrices $\mathbf{C}$ and $\mathbf{D}$. Prove that the matrix multiplication between them *is* commutative.

4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by the matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 2 & -4 & 5 \\ 2 & 0 & 1 \\ 0 & 5 & 2 \end{bmatrix}$$

(a) Specify the domain and codomain.

i. Domain:

ii. Codomain:

(b) What would be the size of the matrix if it were to have a domain of $\mathbb{R}^5$ and a codomain of $\mathbb{R}^2$?

___ $\times$ ___

(c) Write the transformation in the $T(input) = (output)$ format.

(d) What are the *image* and *pre-image* in a transformation?

(e) Find the image of $\vec{x} = (1, 4, 5)$ under the transformation T .

5. Use Cramer's Rule to solve $\mathbf{A}\vec{x} = \vec{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \\ 1 & 5 & 3 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

4 Week 5: Vector Spaces & Subspaces

4.1 Quick Review

From here on out you will be breaking into the core of Linear Algebra (and personally, the most fun part!). Everything from here on out will feel very unfamiliar, but very well within your grasp!

In order to make Linear Algebra applicable (both in the real-world and theoretically), we first have to make some assumptions. There is no way we can *prove* these assumptions, we can only really declare them as true. To indicate that these are assumptions on which everything will hinge upon, we call them **axioms**. For vector spaces, a mathematical “terrain” that contains a *set of vectors* (more on what a *set of vectors* is will be covered in sections 4.3–4.4), we must follow 10 axioms (note that the below is a definition, not a theorem!).

Definition 6: Axioms for a Vector Space

Let V be a nonempty set of objects in which *addition* and *scalar multiplication* are defined. *Addition* of two objects is a *sum*, defined by $\mathbf{u} + \mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ are arbitrary objects in V . *Scalar multiplication* is a process by which we associate an object $k\mathbf{u}$ with $\mathbf{u}$ where k is itself not in V but $k\mathbf{u}$ is.

For the set V to be a **vector space** and $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ to be vectors, the following axioms **MUST** be satisfied:

1. If $\mathbf{u}, \mathbf{v} \in V$, then $(\mathbf{u} + \mathbf{v}) \in V$.
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. There must exist $\mathbf{0} \in V \mid \mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u}$ ($\mathbf{0}$ is called the *zero vector*)
5. $\forall \mathbf{u} \in V \exists -\mathbf{u} \in V \mid \mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$ ($-\mathbf{u}$ is called the *negative* of $\mathbf{u}$)
6. If k is any scalar, then $k\mathbf{u} \in V$
7. $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$
8. $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$
9. $k(m\mathbf{u}) = (km)(\mathbf{u})$
10. $1\mathbf{u} = \mathbf{u}$

As a side note, I was very intentional in putting as much “mathiness” as possible in the above axioms. It is *very* helpful to know how to read these statements and it will make your proofs shorter to write too! (If you want something a bit easier to read, the textbook has the same definitions on page 184)

The two axioms that are bolded (axioms 1 and 6) are key to showing that some vector space is actually a *subspace* of another, larger vector space (for example, $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$). If you know that a proposed vector space W is part of a larger vector space V **AND** that these axioms hold, you have proven that W is itself a vector space.

Linear Independence: A nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ in a vector space V is linearly independent iff

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_r\mathbf{v}_r = \mathbf{0} \quad \text{only when} \quad k_1 = k_2 = \dots = k_r = 0.$$

Span: The span of a nonempty set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is a vector space W consisting of all possible linear combinations of S .

$$W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\} = \text{span}(S)$$

4.2 Problems

1. Below are proposed definitions for a set of objects S , *vector addition*, and *scalar multiplication*. (i) Write three vectors that would satisfy the proposed condition for S , (ii) determine if they will make up a vector space, and (iii) prove that certain axioms hold.

(a) S is the a set of vectors defined by $S = \{(x^2, y^2) \mid x, y \in \mathbb{R}\}$. Vector addition and multiplication is that of ordinary $\mathbb{R}^n$ vector addition and multiplication.

- i. $(\quad, \quad) (\quad, \quad) (\quad, \quad)$
- ii. Is this a valid vector space?
- iii. **(PROOF)**. Prove whether or not axioms 6 and 7 will hold.

(b) Q is the a set of vectors defined by $Q = \{(xy, 0) \mid x, y \in \mathbb{R}\}$. Vector addition and multiplication is that of ordinary $\mathbb{R}^n$ vector addition and multiplication.

- i. $(\quad, \quad) (\quad, \quad) (\quad, \quad)$
- ii. Is this a valid vector space?
- iii. **(PROOF)**. Prove whether or not axioms 3 and 8 will hold.

- iv. Does Q span $\mathbb{R}^2$?
- v. Is Q a linearly independent set of vectors? Phrased another way, can you get the same vector using two different combinations of x and y ?

2. Since vector spaces need to satisfy only 10 axioms (and if they're a subspace, only 2!), there is a lot of freedom in the definition of a vector space. Write an example vector for each specified vector space.

(a) $\mathbb{P}_4$:

(b) $\mathbb{C}_3(-\infty, \infty)$:

(c) $\mathbb{R}_\infty$:

(d) **(BONUS)**. Going off of the number of values associated with the vector spaces from parts (a), (b), and (c), which one is similar to $\mathbb{R}^5$?

3. Let S be a set with a single vector, meaning $S = \{\mathbf{v}\}$.

(a) **(PROOF)**. Assuming that $\mathbf{v} \neq \mathbf{0}$, prove that S is linearly independent.

(b) Now, let $\mathbf{v} = (k, 2k)$, where $k \in \mathbb{R}$. Does S span $\mathbb{R}_2$? Is S a vector space?

(c) Letting $\mathbf{v}$ be the same as defined in part (b) but with the added restriction of $k \neq 0$, is S a vector space? If so, how do you know? If not, why?

4. Let $S = \{(5x^2 + 3x), (4x + 6), (7)\} \in \mathbb{P}_2$.

(a) Write all of the vectors as a three-tuple (a three-tuple is written as (a, b, c)).

$\left(\quad , \quad , \quad \right) \quad \left(\quad , \quad , \quad \right) \quad \left(\quad , \quad , \quad \right)$

(b) Show that this set of vectors is *linearly independent*.

(c) $\mathbf{u} = (3x^2 + 6x + 19)$ is a vector chillin' in $\mathbb{P}_2$. Write $\mathbf{u}$ as a linear combination of the vectors in S .
(Though you haven't learned this yet, S is a *basis* for $\mathbb{P}_2$).

(d) **(BONUS)**. In part (a) you converted a vector in $\mathbb{P}_2$ to a vector in $\mathbb{R}^3$. This is called an *isomorphism*.
Based on this, what do you think an *isomorphism* is?

(e) **(BONUS)**. In part (d) you were introduced to the idea of an *isomorphism*. In general, you cannot use an isomorphism if two vector spaces have different *dimensionalities*. Using this information and the common definition of 'dimensions,' what might the *dimensionality* of $\mathbb{P}_2$ and $\mathbb{R}^3$ be?

5 Week 6: The Vector Basis, Dimension, & Change of Basis

5.1 Quick Review

Even more important than the definition of a vector space, is how we categorize and work with them. This week you will learn just how we do that and even how we move from one vector space to another!

In every math class you have probably discussed the notion of a *coordinate*. But now the question is: *What is a coordinate?* Vectors have coordinates, so it cannot just be a point in space since vectors are not restricted to being just arrows in space.

So, we have to find if there is a systematic way to describe a vector's qualities for *every* vector in a vector space. Recall that the most important properties of a vector space are:

1. Closure under vector addition
2. Closure under scalar multiplication

Using these two properties we can extrapolate that there must be some set of vectors that spans the entirety of the vector space. We call this set of vectors a **basis**. Since every vector v can be described using a basis, then we can associate a quantity of each vector in a basis (a basis vector) all adding together to form v : a linear combination!

So we did it, coordinates are just a linear combination of basis vectors!

Now we can ask some questions about bases:

- What does the number of vectors in a basis space mean?
- How can I get from basis to another basis?
- Can a basis have an amount of vectors that I'd like?
- Is every vector space a basis?

As we work with bases, the answers to these questions should become quite obvious!

Definition 7: Basis

Let V be a vector space. Let $S \in V$ be a finite set of vectors. S is a **basis** for V if:

1. S spans V .
2. S is linearly independent.

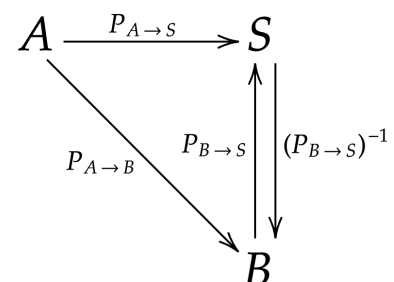
Definition 8: Dimension

Denoted by $\dim(V)$, the dimension is the number of vectors required to fully-define a vector in a vector space. This is also the amount of basis vectors in a vector space's basis.

When changing bases, it is often very helpful to visualize the process with a diagram as shown on the right. In this example

$$P_{A \rightarrow B} = (P_{A \rightarrow S})(P_{B \rightarrow S})^{-1}.$$

This visually demonstrates how we have to move from one basis to another. As you can see to go from A to B , we have to go through the standard basis S .



5.2 Problems

1. A set of objects $S = \{(1, 3, 5), (6, 7, 2), (2, 0, 1)\}$ is a basis for $\mathbb{R}^3$.

- (a) Is S a vector space? _____
- (b) Is S linearly independent? _____
- (c) If S is linearly independent, prove that it is.

- (d) Does S span $\mathbb{R}^3$? _____
- (e) If S spans $\mathbb{R}^3$, prove that it does.

2. You are give a set of vectors $S = \{(3x^3 + 4x), (5x^2 + 3), (3x^3 + x), (x^4 + 5x^2)\} \in \mathbb{P}_4$.

- (a) Is it possible for S to span $\mathbb{P}_4$? If not, why not?

(b) Is S a linearly dependent set of vectors? _____

(c) Based on your answers for parts (a) and (b), is S a basis for $\mathbb{P}_4$? Why?

(d) You should have found that S is *not* a basis. Do something to make S a valid basis:

(e) How did your answer to part (e) affect the dimension of S ?

3. **(PROOF).** Prove that if a set of vectors S is a basis for some vector space V , then S itself is a vector space.

4. Below are three different bases for $\mathbb{R}^3$.

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T = \{(3, 0, 7), (2, 1, 0), (0, 5, 1)\}$$

$$U = \{(3, 0, 0), (0, 2, 6), (4, 5, 0)\}$$

(a) What is the name of S ? _____

(b) Write the transformation matrix for $S \rightarrow T$.

(c) Write the transformation matrix for $S \rightarrow U$.

(d) Write the transformation matrix for $T \rightarrow U$. (Hint: Use the “Ultimate Transformation Triangle!”)

(e) Let $v = (12, 8, -6)$ be defined in terms of the standard basis of $\mathbb{R}^3$. Find $[v]_T$ and $[v]_U$.

5. **(BONUS)**. I promised you guys some applications for linear algebra, so here we go! The information below has been found with the help of <https://courses.cit.cornell.edu/ece4060/Lectures/handout8.pdf> and [Ch 2: What are kets and wavefunctions? | Maths of Quantum Mechanics](#).

Quantum mechanics tells us that some quantities such as the energy $\mathbf{E}$ have a discrete set of values that form the overall quantum state. We can represent this state using an energy basis (a special set of vectors called an *eigenbasis*) $E = \{E_1, E_2, \dots, E_n\}$.

- (a) Write the quantum state ψ_s using the energy basis E with c_i being the coefficients. (Hint: use summation notation!)

- (b) Now imagine we had an *infinitely-sized* set of basis vectors associated with some quantity $\mathbf{X}$. We should also be able to represent the quantum state by forming a linear combination of the continuous and infinite basis $X = \{x_1, x_2, x_3, \dots\}$.

What vector space might the basis X be for? (Such as $\mathbb{R}^7$ or $\mathbb{P}_3$)

- (c) Since X is a set of continuous and infinite functions, how may we denote the quantum state ψ_s now? (Hint: how do we represent the continuous version of summation?)

6 Week 7: The Fundamental Matrix Spaces & Matrix Transformations

6.1 Quick Review

The idea of vector spaces has allowed us to provide an infinite amount of bases for almost any vector space (minus some wacky ones like $S = \{\emptyset\}$; try to think about why!). Now it's time to shift back to thinking about matrices (which are generally vectors) and the information we can extract from them.

Take some time to think about how long it may take for a chromebook to compute a fluid simulation with $500,000 \times 500,000$ elements (that's 2.5×10^{11} elements!). What if we found that we could represent the same system using just 10 vectors and 10 variables (the rest became zeroes!). We'd see a huge performance jump and we've just determined the **rank** of the matrix. The rank tells us how many variables and equations we actually need to describe a system. To understand why rank is crucial consider linear independence; if we had a set of 5 dependent vectors, do we *really* need all 5?

Mathematically (because that's what truly matters), the rank of a matrix is equal to the dimension of its row and column spaces.

Definition 9: Row Space

The row space of a matrix (represented by $\text{row}(A)$) is the space spanned by the row vectors of a matrix. A matrix with more rows (or equations) than the dimension of the row space is said to *overdetermined*.

Definition 10: Column Space

The column space of a matrix (represented by $\text{col}(A)$) is the space spanned by the column vectors of a matrix. A matrix with more columns (or variables) than the number of rows (or equations) in the column space is said to *underdetermined*.

Another key idea is the **null space** of a matrix. You've actually been introduced to this idea, but under a different name: the *kernel*.

A fun way to think about the kernel (and if you ever asked me how to describe it, you'll certainly have heard this because it's a fun explanation), is to imagine squeezing the entire universe into a single point (as shown below). The kernel is the process used to do that.

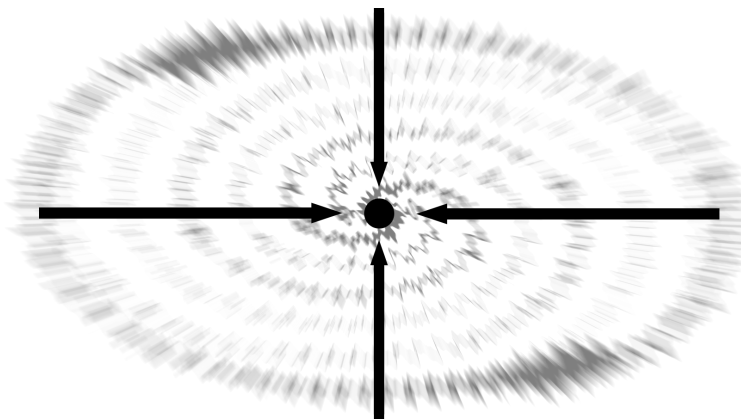


Figure 1: Honestly, I just wanted an excuse to make this.

6.2 Problems

1. The matrix $\mathbf{A}$ has been row-reduced as shown below.

$$\mathbf{A} = \begin{bmatrix} 2 & 7 & 0 & 2 \\ 2 & 0 & 0 & 6 \\ 5 & 3 & 0 & 6 \\ 1 & 0 & 10 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 10 & 3 \\ 0 & 1 & -10 & -3 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -5 \end{bmatrix}$$

- (a) What is the row space of $\mathbf{A}$?
- (b) What is the column space of $\mathbf{A}$?
- (c) What is the null space of $\mathbf{A}$?
- (d) What is the rank of $\mathbf{A}$? _____
- (e) What is the nullity of $\mathbf{A}$? _____
2. Create a matrix $\mathbf{A}$ that has a nullity of 2 and a row space basis with 2 vectors.

3. **(PROOF)**. Prove that if an $n \times n$ matrix $\mathbf{A}$ can be row reduced to become the identity matrix I_n , then the row space takes the form of the standard basis of $\mathbb{R}^n$. Then, prove that $\text{col}(\mathbf{A}) = \text{row}(\mathbf{A})$.

4. Let $T_A : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ be a transformation defined by $T_A(x_1, x_2, x_3) = (2x_1, x_3 + x_2, \frac{1}{2}x_2 + 5x_3)$.

(a) Find the null space of the transformation.

(b) If we were to replace x_1 from the image and replace it with x_2 , what would be the null space then?

- (c) Now let the vectors of T_A be a set of vectors. Is this set of vectors a basis for $\mathbb{P}_2$? Could it be a basis for $\mathbb{R}^3$?

5. Find the transformation matrix T that represents this sequence of transformations applied in $\mathbb{R}^2$:

- i) Rotation counterclockwise by 180° .
- ii) Shear up by 2.
- iii) Compression along the x-axis by a factor of 5.
- iv) Scale by a factor of 5.

6. Using T from question 5:

- (a) Prove that T is invertible.

(b) Find T^{-1}

(c) Apply the transformation T^{-1} to the vector $\vec{v} = (6, 7)$.